**Assignment - Module 1**

1. In which OSI layer the Wi-Fi standard/protocol fits.

   The Wi-Fi standard, defined by the **IEEE 802.11** family, resides strictly within the first two layers of the OSI (Open Systems Interconnection) model: the **Physical Layer (Layer 1)** and the **Data Link Layer (Layer 2)**.

**1. Physical Layer (Layer 1)**

The Physical Layer (PHY) handles the transmission and reception of raw bitstreams over a physical medium—in this case, **Radio Frequency (RF)** waves.

- **Functions:** It defines modulation schemes (like BPSK, QAM), frequency bands (2.4 GHz, 5 GHz, 6 GHz), and channel widths (20 MHz to 320 MHz).
- **Sub-layers:**
  - **PLCP (Physical Layer Convergence Procedure):** Acts as a bridge between the MAC layer and the air, adding a preamble and a header to the data packet to help the receiver synchronize.
  - **PMD (Physical Medium Dependent):** Handles the actual transmission of signals over the air, including coding and modulation.

**2. Data Link Layer (Layer 2)**

Wi-Fi splits the Data Link Layer into two distinct sub-layers to handle hardware-specific communication and high-level data flow.

**Media Access Control (MAC) Sub-layer**

This is the "heart" of Wi-Fi logic. Because wireless is a shared medium where anyone can transmit, the MAC layer manages access to prevent collisions.

- **CSMA/CA:** Wi-Fi uses **Carrier Sense Multiple Access with Collision Avoidance**. Unlike Ethernet (which uses Collision Detection), Wi-Fi devices "listen" to the air and wait for a clear signal before sending to avoid interference.
- **Addressing:** Uses **MAC addresses** (48-bit) to identify specific devices (Source, Destination, and Access Point).
- **Management:** Handles association, authentication, and roaming between Access Points (APs).

**Logical Link Control (LLC) Sub-layer**

The LLC acts as the interface between the Wi-Fi-specific MAC layer and the Network Layer (Layer 3/IP).

- **Function:** It multiplexes protocols (ensuring that an IP packet and an ARP packet can both run over the same Wi-Fi connection) and provides flow control and error checking.

2. Can you share the Wi-Fi devices that you are using day to day life, share that device's wireless capability/properties after connecting to network. Match your device to corresponding Wi-Fi Generations based on properties.

| | |
|---|---|
| SSID: | Hari_5G |
| Protocol: | Wi-Fi 5 (802.11ac) |
| Security type: | WPA2-Personal |
| Manufacturer: | MediaTek, Inc. |
| Description: | MediaTek Wi-Fi 6 MT7921 Wireless LAN Card |
| Driver version: | 3.0.1.1336 |
| Network band (channel): | 5 GHz (44) |
| Aggregated link speed (Receive/ Transmit): | 866/866 (Mbps) |

3. What is BSS and ESS?

**1. BSS (Basic Service Set)**

The **BSS** is the smallest building block of a wireless local area network (WLAN). It consists of a single Access Point (AP) and all the wireless devices (stations or STAs) associated with it.

- **BSSID (Basic Service Set Identifier):** This is the unique hardware identifier for the BSS. It is typically the **MAC address** of the Access Point's wireless radio.
- **Infrastructure Mode:** In this setup, all communication must pass through the AP. If Device A wants to send data to Device B within the same BSS, the data goes from Device A to the AP, and then the AP forwards it to Device B.
- **Coverage Area:** The range of the BSS is limited to the signal reach of that specific AP.

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**2. ESS (Extended Service Set)**

An **ESS** is a collection of two or more interconnected BSSs that appear as a single, seamless network to the user. This is how large-scale Wi-Fi (like in offices, airports, or universities) is constructed.

- **SSID (Service Set Identifier):** In an ESS, all participating Access Points share the **same SSID** (network name). This allows a user to see one network (e.g., "University_WiFi") even though there are hundreds of APs.
- **Distribution System (DS):** This is the backbone (usually wired Ethernet) that connects the various APs in the ESS. It allows data to flow between different BSSs and out to the internet.

- **Roaming:** The primary benefit of an ESS is roaming. A user can move from the coverage area of AP1 to AP2 without losing their connection, because the ESS manages the "handover" between BSSs behind the scenes.

4. What are the basic functionalities of Wi-Fi Accesspoint?

   **1. Beaconing and Network Advertisement**

   The AP periodically transmits a **Beacon Frame** (typically every 100ms). This frame announces the presence of the network to nearby devices.

- **Information included:** The SSID (Network Name), supported data rates, security requirements (e.g., WPA3), and synchronization timestamps.
- **Purpose:** It allows devices to "discover" the network and stay synchronized with the AP's clock.

   **2. Association and Authentication Management**

   Before a device (Station/STA) can send data, it must go through a formal handshake managed by the AP.

- **Authentication:** The AP verifies the identity of the device (via a password or certificate).
- **Association:** Once authenticated, the AP allocates resources and registers the device's MAC address in its association table. This officially makes the device part of the **BSS (Basic Service Set)**.

   **3. Bridging: Wired to Wireless Conversion**

   The AP acts as a Layer 2 bridge. Its primary job is to take **802.3 Ethernet frames** from the wired backbone and convert them into **802.11 Wi-Fi frames** for the air, and vice versa.

- It maintains a bridge table to track which MAC addresses are currently associated with its wireless radio.

   **4. Media Access Control (Traffic Coordination)**

   Since the air is a shared medium, the AP helps manage who speaks and when to avoid "collisions" (multiple devices transmitting at once).

- **CSMA/CA:** It enforces the rules of "listen before you speak."
- **Buffer Management:** If a device is in power-save mode, the AP buffers (stores) the incoming data for that device and notifies it in the next beacon that it has data waiting.

   **5. Security and Encryption**

   The AP is the first line of defense for the network.

- **Encryption:** It handles the encryption/decryption of data packets using protocols like **AES** (Advanced Encryption Standard).

- **Key Management:** It manages the four-way handshake used to derive unique encryption keys for every connected device.

**6. Radio Resource Management (RRM)**

Modern APs intelligently manage the physical radio environment.

- **Channel Selection:** Selecting the least congested frequency channel (e.g., choosing Channel 6 instead of 1 if 1 is crowded).
- **Transmit Power Control:** Adjusting signal strength to provide enough coverage without interfering with neighboring APs.

5. Difference between Bridge mode and Repeater mode.

**1. Bridge Mode**

**Bridge Mode** is used to connect two separate network segments (usually a wired segment and a wireless segment) at **Layer 2 (Data Link Layer)**. In this mode, the Access Point acts as a transparent bridge, passing traffic based on MAC addresses without acting as a router.

- **How it works:** The AP disables its routing functions (DHCP, NAT, Firewall) and simply passes data through. It connects a wireless client (or a remote wired switch) to a primary wired network as if they were on the same physical wire.
- **Key Characteristic:** It creates a single flat network. All devices, whether connected to the Bridge or the main Router, receive IP addresses from the same pool.

**Example:**

Imagine you have a primary router in your living room and a desktop PC in a distant bedroom that lacks a Wi-Fi card. You can place a second AP in the bedroom in **Bridge Mode**. You connect the PC to the AP via an Ethernet cable. The AP "bridges" that Ethernet connection over Wi-Fi to the main router. The PC thinks it is plugged directly into the main router.

**2. Repeater Mode (Range Extender)**

**Repeater Mode** is used to extend the physical coverage area of an existing wireless signal. It functions by receiving a wireless signal from a base AP and re-broadcasting it to a new area.

- **How it works:** The device has one radio that talks to the "Root AP" and another (or the same one shared) that talks to local clients. It effectively "repeats" every packet it hears.
- **Key Characteristic:** It often results in a **50% reduction in throughput** because a single-radio repeater cannot receive and transmit at the same time; it must alternate between talking to the router and talking to the client.

**Example:**

Your router is at one end of a long hallway, and the signal is weak in the kitchen at the other end. You place a device in **Repeater Mode** halfway down the hall. It picks up the signal from the router and blasts it into the kitchen. Your phone in the kitchen connects to the Repeater, which then passes the data back to the router.

**Detailed Comparison Table**

| Feature | Bridge Mode | Repeater Mode |
|---|---|---|
| **Primary Goal** | Connects two different network types (Wired to Wireless). | Extends the range of an existing Wireless signal. |
| **Network Layout** | Usually connects a wired device to a wireless network. | Connects wireless devices to a wireless network. |
| **Performance** | High; preserves the speed of the wireless link. | Lower; throughput is typically cut in half (Latency increases). |
| **Routing/NAT** | Disabled; the device is transparent. | Disabled; it just passes packets along. |
| **IP Addressing** | IP addresses come from the main router. | IP addresses come from the main router. |
| **Hardware Use** | Connects to a switch or PC via Ethernet. | Placed in a "dead zone" to boost signal. |

6. What are the differences between 802.11a and 802.11b.

**Frequency Band and Interference**

- **802.11a:** Operates in the **5 GHz** band. This band was much "cleaner" at the time because few consumer devices used it.
- **802.11b:** Operates in the **2.4 GHz** band. This is a very crowded frequency shared by microwave ovens, cordless phones, and Bluetooth devices, making it highly susceptible to interference.

**2. Maximum Data Rate (Speed)**

- **802.11a:** Offered significantly higher speeds for its time, topping out at **54 Mbps**.
- **802.11b:** Was much slower, with a maximum data rate of only **11 Mbps**.

**3. Range and Penetration**

- **802.11a:** Due to the higher frequency (5 GHz), it has a **shorter range** and struggles to penetrate solid objects like walls and floors.
- **802.11b:** The lower frequency (2.4 GHz) provides a **longer range** and better penetration through obstacles, making it more reliable for covering larger homes.

**4. Modulation Technique**

- **802.11a:** Uses **OFDM** (Orthogonal Frequency Division Multiplexing). This is a more efficient method of encoding data that splits the signal into multiple smaller sub-signals to reduce interference.
- **802.11b:** Uses **DSSS** (Direct-Sequence Spread Spectrum) with **CCK** (Complementary Code Keying). This is a simpler, older modulation technique that is less efficient than OFDM.

7. Configure your modem/hotspot to operate only in 2.4GHz and connect your laptop/Wi-Fi device, and capture the capability/properties in your Wi-Fi device. Repeat the same in 5GHz and tabulate all the differences you observed during this.
8. What is the difference between IEEE and WFA?

**1. IEEE (Institute of Electrical and Electronics Engineers)**

The IEEE is a global professional organization that creates the technical **standards** for a vast array of technologies, including the **802.11** working group which defines Wi-Fi.

- **Role:** The **Architect / Engineer**.
- **Primary Function:** They define the mathematical and electrical specifications of the technology. They decide how the radio waves should be modulated, how packets are structured, and how the hardware should behave at a theoretical level.
- **Naming Convention:** They use technical numbers like **802.11b, 802.11n, 802.11ax, or 802.11be**.
- **Focus:** Technical innovation, spectrum efficiency, and physical/data-link layer logic.

**2. WFA (Wi-Fi Alliance)**

The Wi-Fi Alliance is a non-profit **industry trade association** made up of companies (like Apple, Samsung, Intel, and Cisco). It was formed because, in the early days, devices from different manufacturers following the same IEEE standard often couldn't talk to each other.

- **Role:** The **Marketer / Certifier**.
- **Primary Function:** They ensure **interoperability**. They take the IEEE standards and create a certification program. If a device passes their rigorous testing, it earns the right to display the "Wi-Fi CERTIFIED" logo.
- **Naming Convention:** To make it easier for consumers, they rebranded the IEEE numbers into simplified generations like **Wi-Fi 4, Wi-Fi 5, Wi-Fi 6, and Wi-Fi 7**.
- **Focus:** Consumer experience, branding, and ensuring a "plug-and-play" ecosystem.

9. List down the type of Wi-Fi internet connectivity backhaul, share your home/college's wireless internet connectivity backhaul name and its properties.

**Types of Wi-Fi Internet Connectivity Backhaul**

1. **Wired Backhaul (Most Common/Reliable)**
    - **Ethernet (Copper):** Standard Cat5e/Cat6 cables connecting APs to a local switch. Supports speeds up to 10 Gbps.
    - **Fiber Optics:** Uses light to transmit data. Essential for long distances (between buildings) and ultra-high bandwidth (100 Gbps+).
    - **DSL/Cable:** Common in residential setups where the Wi-Fi router connects to the ISP via a telephone line (DSL) or coaxial cable (DOCSIS).
2. **Wireless Backhaul**
    - **Wi-Fi Mesh Backhaul:** In mesh systems, one radio band (usually a dedicated 5GHz or 6GHz band) is used exclusively for AP-to-AP communication.
    - **Point-to-Point (PtP) Microwave:** High-gain directional antennas connect two locations (up to several miles apart) wirelessly. Used when laying cable is too expensive.
    - **Satellite Backhaul:** Used in remote areas or moving vehicles (e.g., Starlink). Data travels from the AP to a satellite dish, then to space.
    - **Cellular (4G/5G) Backhaul:** Mobile hotspots or industrial routers use a SIM card to provide internet via cellular towers.

**Backhaul Type: Enterprise Fiber-Optic Backhaul**

- **Physical Medium:** Single-Mode Fiber (SMF).
- **Protocol:** 10GBase-SR or 40GBase-LR.
- **Properties:**
    - **Extreme Bandwidth:** Capable of handling thousands of simultaneous users (students/faculty) without congestion.
    - **Low Latency:** Fiber provides the lowest possible delay (ping), which is critical for video conferencing and research applications.
    - **High Reliability:** Fiber is immune to electromagnetic interference (EMI) from power lines or other radio signals, ensuring a stable connection.
    - **Scalability:** The university can increase speed (e.g., from 10 Gbps to 100 Gbps) by simply changing the transceivers (SFP modules) at each end without digging up the ground to replace the cable.

10. List down the Wi-Fi topologies and use cases of each one.

**1. Independent Basic Service Set (IBSS) / Ad-Hoc Mode**

In an IBSS topology, devices connect directly to one another without the use of an Access Point (AP). It is a peer-to-peer (P2P) network where every station (STA) handles its own management and synchronization.

- **How it works:** Devices within radio range "negotiate" a connection. There is no central hub or connection to a wired backbone.
- **Use Cases:** * **Quick Data Transfer:** Two people in a remote area needing to transfer a file between laptops.
    - **Emergency Services:** First responders setting up a temporary network in a disaster zone where infrastructure is destroyed.
    - **Legacy Gaming:** Local handheld gaming (e.g., older Nintendo DS/PSP consoles).

### 2. Infrastructure Mode (BSS & ESS)

This is the most common topology. It relies on a central Access Point (AP) to manage all communications.

- **Basic Service Set (BSS):** A single AP and its associated clients. All traffic goes through the AP.
- **Extended Service Set (ESS):** Multiple APs connected via a wired distribution system (Ethernet) to provide a large coverage area.
- **Use Cases:**
    - **Residential:** Home Wi-Fi routers.
    - **Enterprise:** Office buildings, airports, and university campuses where seamless roaming is required.

### 3. Mesh Topology (802.11s)

In a Mesh network, multiple Access Points (called Mesh Points) connect to each other wirelessly to route data across the network. Only one or two nodes need to be physically plugged into a wired internet source.

- **How it works:** Nodes use a "self-healing" pathfinding algorithm. If one node fails, the data is automatically rerouted through another available node.
- **Use Cases:**
    - **Smart Homes:** Providing coverage in large houses where running Ethernet cables to every room is difficult.
    - **Outdoor/Municipal Wi-Fi:** Covering city parks or large outdoor events where wiring is impractical.